

How Paper Displays Work

The Legacy

The idea of paper displays, or ePaper as it's also known, is hardly new. Back in 1975, Nicholas Sheridan, a physicist working at the Xerox PARC (Palo Alto Research Center), started his research on ePaper. The novel idea back then was to replace paper with portable flat-panel displays, which, as we all know, were not exactly portable back then, were very expensive, and had terrible contrast—you could hardly make out the screen in a lit room, forget about outdoors!

Sheridan's dream was to make a flexible paper-like sheet that could not only display text and images, but also be re-written by using an electrostatic charge. The solution was to use tiny electrically-charged spheres, enclosed in the oil-filled cavities of a thin layer of a transparent elastomer.

Incidentally, this is the technology Xerox PARC still uses in its research of ePaper, which they call SmartPaper. Each 100-micron sphere is coloured half white and half black (or another dark colour). The spheres contain a dipole—like a magnet, with positive and negative charges on opposite ends.

Thus, when an electric charge is passed over certain pre-designated areas of the elastomer film, the

spheres in those areas rotate to show their dark side, colouring the area. A chip controls charges to make an electronically-controlled etch-a-sketch!

Since the spheres are reflective in nature, they are extremely high-contrast, making them readable even under direct sunlight. This technology has already been used to make some interesting products. We will talk about those a little later. Needless to say, this technology, though easily implementable, has the drawback of being monochrome. Still, it's a step in the right direction.

The Leaders

Say the words "paper display" and the name that springs to mind is E-Ink Corporation—the Xerox or Discman of paper displays!

E-Ink currently provides technology and paper displays to many corporates—not just to help them build prototypes, but also for full-fledged products. E-Ink's investors include Toppan Printing of Japan, Intel Capital Corporation, Motorola Corporation, Philips Compo-nents (a division of Royal Philips Electronics) and Vivendi Univer-sal Publishing (France).

The list of companies that are actively researching paper displays